

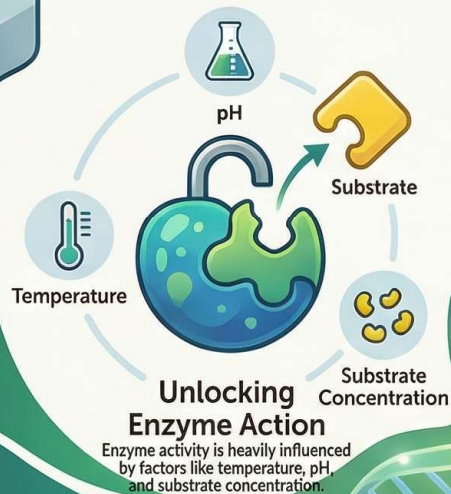
Core Techniques in Biotechnology: From Analysis to Engineering

Measuring with Light:
Spectrophotometry



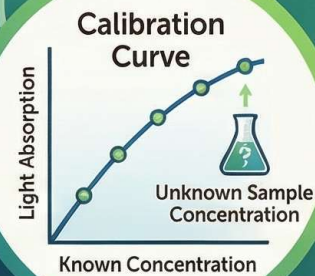
Biochemical Principles & Analysis

This technique measures a substance's concentration by how much light it absorbs.



Unlocking Enzyme Action

Enzyme activity is heavily influenced by factors like temperature, pH, and substrate concentration.



Creating a "Ruler": The Calibration Curve

Plotting known concentrations against light absorption allows for determining unknown sample concentrations.

Genetic Engineering in Practice

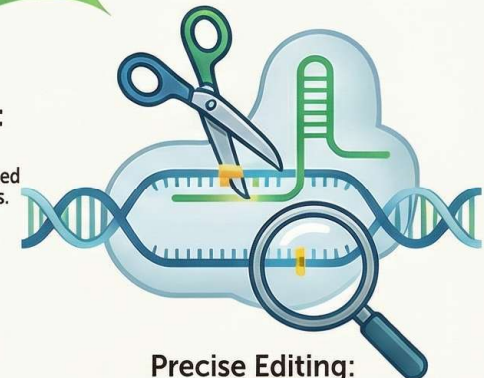
Bacterial Transformation: Inserting New DNA

Plasmids (small, circular DNA) are introduced into bacteria using a "heat shock" process.



Making Changes Visible: Reporter Genes

Genes like GFP (Green Fluorescent Protein) provide a visual signal, like a green glow.



Precise Editing: The CRISPR-Cas9 System

This system acts like molecular scissors to cut DNA at a specific, targeted location.